SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

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Annexure A

Debugging & Optimisation Task

Code of Conduct:

I Maniyam Anushka(192525411) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Maniyam Anushka

Annexure A

Debugging & Optimisation Task

1. A rapidly growing IT company has shifted its operations to a newly constructed six-storey smart office building. More than **1,200 employees** use wireless laptops, smartphones, IP phones, printers, and IoT devices connected through the company's **Wireless Local Area Network (WLAN)**. The office relies heavily on cloud computing, online meetings, video conferencing, and file sharing, making uninterrupted wireless connectivity essential for daily operations. In addition, the company has a team of field engineers and sales executives who travel across different cities and use **4G/5G Wireless Wide Area Network (WWAN)** services to access the company's servers, customer databases, and business applications from remote locations.

   During the last few weeks, employees have reported several connectivity issues. Users experience slow internet speeds during peak working hours, frequent Wi-Fi disconnections while moving between floors, and weak wireless signals in conference rooms. Video conferencing applications often freeze or disconnect, affecting communication with international clients. Network administrators have also observed that some wireless access points are overloaded while others remain underutilized, leading to uneven traffic distribution. Furthermore, field engineers working in rural and highway areas frequently lose WWAN connectivity, delaying the upload of maintenance reports and customer information. Security audits reveal that a few wireless devices are still connected using outdated security protocols, exposing sensitive company information to potential cyber threats. The management has requested the network engineering team to investigate these issues, identify the root causes, and optimize both the WLAN and WWAN infrastructure to ensure high performance, secure communication, and seamless connectivity for all users.

2. A multinational financial organization operates its headquarters, regional offices, and multiple data centers using an **Asynchronous Transfer Mode (ATM) network** to support mission-critical services such as online banking, video conferencing, Voice over IP (VoIP), and real-time financial transactions. The ATM network uses fixed-size **53-byte cells** and establishes **Virtual Circuits (VCs)** to provide reliable and predictable Quality of Service (QoS). Recently, the organization has experienced several performance issues during peak business hours. Customers report delays in online transactions, voice calls suffer from jitter and occasional packet loss, and video conferencing sessions experience reduced quality. Network monitoring indicates that several ATM switches are operating at high utilization, some Virtual Circuits are congested, and available bandwidth is not being efficiently allocated among different traffic classes. Engineers also discover that certain applications are using inappropriate ATM service categories, leading to increased delay and inefficient resource utilization. The IT department has assigned you as the network engineer to investigate the ATM network, identify the faults, debug the performance issues, and recommend optimization strategies to restore efficient, reliable, and high-quality communication across the enterprise.
3. A multinational e-commerce company has interconnected its headquarters, regional offices, warehouses, and cloud data centers using a **Multiprotocol Label Switching (MPLS)** network to ensure fast, secure, and reliable communication. The MPLS infrastructure supports critical business applications, including online order processing, enterprise resource planning (ERP), video conferencing, Voice over IP (VoIP), cloud-based inventory management, and real-time

customer support. Recently, the organization has experienced frequent performance issues during peak business hours. Employees report slow access to cloud applications, delayed order processing, poor-quality voice calls, and interruptions during video conferences. Network monitoring reveals that some Label Switched Paths (LSPs) are heavily congested, while other paths remain underutilized. The IT team also discovers that several routers have incorrect label mappings, causing inefficient routing and increased latency. In addition, traffic engineering has not been properly configured, resulting in uneven bandwidth utilization across the MPLS backbone. Since these issues are affecting customer satisfaction and business operations, the network administrator has asked you to investigate the MPLS network, identify the faults, debug the performance problems, and recommend suitable optimization techniques to improve the overall efficiency and reliability of the network.

4. A multinational logistics company operates an automated warehouse where barcode scanners, RFID readers, autonomous robots, and inventory management systems continuously exchange data over a computer network. Every day, thousands of shipment records, customer details, and inventory updates are transmitted between warehouse terminals and the central server. Recently, the company has observed that some shipment records are arriving with incorrect information, barcode scans occasionally fail, and inventory quantities in the database do not always match the physical stock. During peak hours, network monitoring reports an increase in transmission errors caused by electrical interference, signal attenuation, and network congestion. Although some corrupted frames are detected and retransmitted, others remain undetected, resulting in incorrect database entries and delayed order processing. The IT department suspects that the current error detection and correction mechanisms are not configured efficiently. As the network engineer, you are assigned to investigate the transmission errors, identify the weaknesses in the existing error control techniques, and recommend suitable debugging and optimization strategies to improve the reliability and accuracy of data communication.
5. A leading online education platform conducts live classes and online examinations for more than **15,000 students** across different locations. During examinations, the central server simultaneously sends question papers, images, videos, and answer submission acknowledgements to all students over the network. The server is equipped with a high-speed **10 Gbps** connection, whereas many students access the system using slower broadband or mobile internet connections. During peak hours, students experience incomplete downloads, delayed page loading, duplicate packets, and repeated retransmissions. Some students receive the next set of data before their devices finish processing the previous one, causing receiver buffer overflow and application crashes. Network administrators also observe an excessive number of retransmissions, increased latency, reduced throughput, and poor bandwidth utilization. The current flow control mechanism is not effectively regulating the transmission rate between the sender and receivers. As the network engineer, you have been assigned to investigate the communication problems, identify the causes of inefficient flow control, debug the existing system, and recommend suitable optimization techniques to ensure reliable and efficient data transmission during high-traffic periods.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

SCENARIO 1: Enterprise WLAN & WWAN Connectivity Optimization

2. Aim

To investigate, debug, and optimize the Wireless Local Area Network (WLAN) and Wireless Wide Area Network (WWAN) infrastructure of a six-storey smart office so that all 1,200+ employees and remote field engineers get reliable, secure, and high-speed wireless connectivity.

3. Problem Statement

A growing IT company's six-storey smart office uses WLAN for laptops, IP phones, printers and IoT devices, and WWAN (4G/5G) for field staff. Employees face slow speeds at peak hours, Wi-Fi drops while moving between floors, weak signal in conference rooms, video-call freezing, uneven Access Point (AP) load, WWAN dead zones on highways, and devices still using outdated, insecure Wi-Fi protocols. The network team must find the root causes and optimize both networks.

4. Network Overview

WLAN (Wireless Local Area Network) connects devices inside a limited area, such as one building, using IEEE 802.11 standards (Wi-Fi) and Access Points (APs) linked to a wired backbone. WWAN (Wireless Wide Area Network) connects mobile users over a large geographic area using cellular technologies such as 4G LTE and 5G, provided by telecom operators. In this office, every floor has multiple APs connected to floor switches, which uplink to a core switch and then to the internet gateway and cloud services. Field engineers rely on WWAN through carrier towers to reach the same corporate servers via VPN.

5. Problem Identification

5.1 Faults, Root Causes and Impact

Fault Observed	Root Cause	Impact on Network
Slow internet during peak hours	Too many devices sharing limited channel bandwidth; no bandwidth prioritization (QoS)	Poor user experience, delayed file sync, slow cloud access
Frequent disconnections while moving between floors	No seamless roaming (missing 802.11r/k/v); each floor AP uses a different SSID/controller island	Dropped VoIP/video calls, re-authentication delay
Weak signal in conference rooms	AP placement not based on site survey; signal attenuation from concrete walls and glass partitions	Low RSSI, high retransmissions, call drops

Fault Observed	Root Cause	Impact on Network
Video conferencing freezes	Co-channel interference from overlapping channels (1,6,11 not properly reused); insufficient bandwidth allocation for real-time traffic	Jitter, packet loss, poor audio/video quality
Some APs overloaded, others underused	Static/default AP power and channel settings; no load balancing or band-steering (2.4 GHz vs 5 GHz)	Congestion at hotspots, wasted capacity elsewhere
Field engineers lose WWAN in rural/highway areas	Weak cellular tower coverage, single-carrier dependency, no failover	Delayed report upload, broken VPN sessions
Devices using outdated security protocols	Legacy devices still configured with WEP/WPA-TKIP instead of WPA3/WPA2-AES	Vulnerable to eavesdropping, deauthentication and brute-force attacks

6. Debugging Process

6.1 Step-by-Step Troubleshooting Procedure

Step 1 — Baseline Site Survey: Perform a Wi-Fi site survey using a tool such as Ekahau, NetSpot, or Cisco Prime Infrastructure heatmaps to record Received Signal Strength Indicator (RSSI), noise floor and channel usage on every floor. This is done first because you cannot fix what you have not measured.

Step 2 — Check AP Load and Client Count: Use the WLAN controller dashboard (e.g., Cisco WLC, Aruba Central) or the command 'show wireless client summary' to see how many clients are on each AP. This step reveals overloaded versus underused APs.

Step 3 — Analyze Channel and Interference: Run a spectrum analysis (Wi-Fi Analyzer / AirMagnet) to detect overlapping channels and non-Wi-Fi interference (microwave ovens, Bluetooth). This step is performed to confirm co-channel interference as the cause of freezing calls.

Step 4 — Verify Roaming Configuration: Check whether 802.11k (neighbor report), 802.11v (BSS transition) and 802.11r (fast transition) are enabled on the controller using 'show run wlan <id>'. This confirms whether slow roaming is causing floor-to-floor drops.

Step 5 — Packet Capture on Video Traffic: Use Wireshark on a test laptop during a video call to check for retransmissions, jitter and packet loss values. This quantifies the video freezing complaint with real numbers.

Step 6 — Audit Security Settings: Run 'show wlan security' or check the controller's SSID security tab to list every SSID's authentication method and flag any WEP/WPA-TKIP SSID still active. This step is essential to close the cyber-security gap.

Step 7 — WWAN Signal Audit: Field engineers use a cellular signal-mapping app (e.g., OpenSignal) along their travel routes to log RSRP/RSRQ values and identify dead zones. This step localizes exactly where WWAN coverage fails.

Step 8 — Bandwidth Utilization Check: Use SNMP-based monitoring (PRTG, SolarWinds) to plot uplink utilization on floor switches and the core switch during peak hours, confirming whether the bottleneck is wireless or wired backhaul.

7. Optimization Techniques

7.1 WLAN Optimizations

Technique	Explanation	Advantage	Limitation
AP placement redesign using site survey data	Reposition/add APs so every area has RSSI ≥ -67 dBm with 15-20% cell overlap	Removes dead zones and weak-signal rooms	Requires physical re-cabling or new AP purchase
Channel planning (1/6/11 reuse) + auto RF (RRM)	Controller auto-assigns non-overlapping channels and transmit power	Reduces co-channel interference automatically	Needs a capable WLC/cloud controller
Band steering & load balancing	Push dual-band clients to 5 GHz/6 GHz and move clients from busy AP to nearby idle AP	Balances client load, uses more spectrum	Older single-band client devices cannot move
802.11k/v/r fast roaming	Provides neighbor lists and fast key exchange so client hands off in <50 ms	Seamless roaming between floors, no VoIP drop	All APs and clients must support the standard
QoS / WMM traffic prioritization	Classify VoIP and video into higher priority queues (DSCP EF/AF41)	Video calls stay smooth even under load	Misconfigured QoS can starve normal data traffic
Upgrade to WPA3-Enterprise / 802.1X	Replace WEP/WPA-TKIP with WPA3-SAE and RADIUS-based per-user authentication	Strong encryption, prevents credential attacks	Legacy IoT devices may not support WPA3

7.2 WWAN Optimizations

Deploy dual-SIM/multi-carrier routers in field vehicles so the device automatically fails over to the strongest available carrier. Use store-and-forward (offline caching) in the field app so reports queue locally and sync automatically once signal returns, instead of failing outright. Enable SD-WAN with WWAN as a backup link so head-office connectivity has automatic failover. Compared with the current single-carrier, always-online-required design, this optimized approach trades a small additional router/subscription cost for near-zero data loss in coverage gaps.

8. Network Diagram

```
                     INTERNET / CLOUD SERVICES
                               |
                       [ Firewall / Router ]
                               |
                       [   Core Switch    ]
              _________________|__________________
             |            |             |            |
      [Floor6 SW]  [Floor5 SW]   [Floor2-4 SW]  [Floor1 SW]
             |            |             |            |
     (AP)(AP)(AP)  (AP)(AP)(AP)   (AP)(AP)(AP)  (AP)(AP)(AP)
      Laptops       IP Phones      IoT Sensors    Reception
      Smartphones   Printers       Cameras        Guests

    Remote Field Engineers  --(4G/5G Tower)--> [ISP Core] --> Firewall
    (Laptops/Tablets over VPN, WWAN)
```

9. Flowchart — Troubleshooting Process

```
    [ Start: Complaint Received ]
                |
                v
    [ Site Survey: Measure RSSI/Noise ]
                |
                v
     < Signal Weak in area? >-- Yes --> [ Add/Reposition AP ]
                | No
                v
    [ Check AP Client Load ]
                |
     < AP overloaded? >-- Yes --> [ Enable Load Balancing/Band Steering ]
                | No
                v
    [ Spectrum/Channel Analysis ]
                |
     < Interference found? >-- Yes --> [ Re-plan Channels / Power (RRM) ]
                | No
                v
    [ Check Roaming Config (11k/v/r) ]
                |
     < Roaming disabled? >-- Yes --> [ Enable Fast Roaming ]
                | No
                v
    [ Audit SSID Security ]
                |
     < Weak protocol found? >-- Yes --> [ Migrate to WPA3/802.1X ]
                | No
                v
    [ Validate with Test Calls & Wireshark ]
                |
                v
    [ End: Document & Monitor ]
```

10. Algorithm / Procedure for Optimization

Step 1: Collect baseline RSSI, throughput, jitter and packet-loss data on every floor.

Step 2: Identify APs with RSSI below -67 dBm or client count above vendor-recommended limit (e.g., >30 clients).

Step 3: Re-plan AP placement and non-overlapping channel map (1, 6, 11 on 2.4 GHz; wider set on 5 GHz).

Step 4: Enable RRM (Radio Resource Management) for automatic power/channel tuning.

Step 5: Enable 802.11k/v/r for fast roaming across floor APs.

Step 6: Apply QoS policy: mark VoIP as EF, video as AF41, data as best effort.

Step 7: Reconfigure all SSIDs to WPA3-Enterprise/802.1X; disable legacy WEP/TKIP SSIDs.

Step 8: For WWAN, configure dual-carrier failover and store-and-forward caching on field devices.

Step 9: Re-test with Wireshark/iPerf and compare against baseline; repeat tuning until targets are met.

11. Performance Analysis

Metric	Before Optimization	After Optimization	Expected Improvement
Average throughput per user	2-4 Mbps at peak	15-25 Mbps at peak	Significant increase
Latency (video call)	150-300 ms, spikes to 600 ms	20-60 ms stable	Major reduction
Packet loss	3-6%	<1%	Reduced call/video freezing
AP channel utilization balance	Uneven (some APs >90%, others <20%)	Balanced 40-60% across APs	Better load distribution
Roaming handoff time	800 ms - 2 s (visible drop)	<50 ms (seamless)	No call drops while moving
WWAN coverage gaps	Frequent dead zones on routes	Automatic carrier failover	Continuous connectivity
Security posture	Mixed WEP/WPA-TKIP SSIDs	WPA3-Enterprise/802.1X everywhere	Strong encryption, per-user auth
Scalability (device growth)	Fixed capacity, saturates fast	RRM adapts to added APs/clients	Easily scales with growth

12. Security Improvements

12.1 Risks

Legacy WEP/WPA-TKIP encryption is crackable within minutes using freely available tools; shared PSK passwords cannot identify individual users; unpatched IoT devices can be used as an entry point for lateral attacks; rogue APs can impersonate the corporate SSID (evil twin attack).

12.2 Solutions

Migrate every SSID to WPA3-Enterprise with 802.1X/RADIUS authentication so each employee uses unique credentials. Segment IoT devices onto a separate VLAN with restricted internet access. Deploy Wireless Intrusion Prevention System (WIPS) to detect rogue APs and deauthentication attacks. Enforce certificate-based VPN for all WWAN field connections.

12.3 Best Practices

Regularly patch AP firmware; disable WPS; use MAC-based device onboarding for IoT; conduct quarterly wireless security audits; enforce strong RADIUS server password policies.

13. Testing and Validation

Test Case	Procedure	Expected Output	Result
TC1: Peak-hour throughput	Run iPerf between client and server during peak hours on 3 floors	> 15 Mbps per user	Pass
TC2: Floor-to-floor roaming	Walk test with active VoIP call from Floor 1 to Floor 6	No call drop, handoff < 50 ms	Pass
TC3: Conference room signal	Measure RSSI in all conference rooms	RSSI ≥ -67 dBm	Pass
TC4: Video conference quality	10-minute video call under 50-user load	Jitter < 30 ms, loss < 1%	Pass
TC5: Weak-protocol scan	Run WIPS scan for WEP/TKIP SSIDs	Zero legacy SSIDs found	Pass
TC6: WWAN failover	Disconnect primary carrier signal in field test	Automatic switch to secondary carrier < 10 s	Pass

14. Advantages

Seamless roaming and higher throughput improve employee productivity; QoS ensures reliable video conferencing with international clients; WPA3 migration significantly reduces cyber-security risk; WWAN failover keeps field engineers connected even in weak-coverage zones; RRM-based design scales automatically as the company grows.

15. Limitations

Optimization requires upfront investment in site survey tools, additional APs and a capable WLAN controller; legacy IoT devices without WPA3 support may need replacement; WWAN performance still depends on the external telecom operator's tower coverage, which the company cannot fully control.

16. Future Enhancements

Adopt Wi-Fi 6E/7 for higher capacity and lower latency; integrate AI-driven network analytics for predictive fault detection; extend SD-WAN to dynamically blend WWAN and satellite (LEO) links for field engineers in truly remote areas; move toward a Zero Trust Network Access (ZTNA) model instead of traditional VPN.

17. Conclusion

The connectivity issues in the smart office stemmed from poor AP placement, lack of load balancing, weak roaming support, and outdated security protocols, while field WWAN issues arose from single-carrier dependency. By applying a structured debugging process — site survey, load analysis, spectrum analysis, roaming and security audit — followed by optimization through RRM, QoS, fast roaming, WPA3-Enterprise and multi-carrier WWAN failover, the network achieves higher throughput, lower latency, seamless mobility, and stronger security, meeting the operational needs of a 1,200+ user smart office.

19. References (IEEE Format)

[1] IEEE Std 802.11-2020, "IEEE Standard for Information Technology—Telecommunications and Information Exchange between Systems—Local and Metropolitan Area Networks—Specific Requirements Part 11: WLAN MAC and PHY Specifications," IEEE, 2021.

[2] B. A. Forouzan, Data Communications and Networking, 5th ed. New York, NY, USA: McGraw-Hill, 2013.

[3] Cisco Systems, "Wireless LAN Controller (WLC) Configuration Best Practices," Cisco Documentation, 2023.

[4] Wi-Fi Alliance, "WPA3 Specification," Wi-Fi Alliance, 2020.

[5] 3GPP, "System Architecture for the 5G System," 3GPP TS 23.501, 2022.

SCENARIO 2: ATM Network Performance Debugging & Optimization

2. Aim

To debug and optimize an Asynchronous Transfer Mode (ATM) network used by a multinational financial organization so that online banking, VoIP and video conferencing achieve reliable, low-delay, high-quality performance during peak business hours.

3. Problem Statement

The organization's headquarters, regional offices and data centers are connected using ATM, a cell-switching technology using fixed 53-byte cells and Virtual Circuits (VCs) for guaranteed QoS. During peak hours, transactions are delayed, VoIP suffers jitter and packet loss, and video conferencing quality drops. Some switches show high utilization, VCs are congested, bandwidth is inefficiently allocated, and some applications use the wrong ATM service category. The task is to find root causes and optimize the ATM network.

4. Network Overview

ATM (Asynchronous Transfer Mode) is a connection-oriented switching technology that breaks all traffic — data, voice, video — into fixed-size 53-byte cells (5-byte header + 48-byte payload). Because the cell size is fixed, switching hardware can forward cells at very high, predictable speed. ATM establishes a Virtual Circuit (VC) — either a Permanent Virtual Circuit (PVC) or Switched Virtual Circuit (SVC) — before any data is sent, and each VC is assigned a Service Category (CBR, VBR, ABR, or UBR) that defines its QoS guarantees. In this bank, ATM switches connect the headquarters, regional branch offices, and data centers, carrying core banking transactions (needs low delay), VoIP (needs constant low jitter), and video conferencing (needs guaranteed bandwidth) over the same backbone.

5. Problem Identification

5.1 ATM Service Categories (background needed before fault analysis)

Service Category	Meaning	Typical Use
CBR (Constant Bit Rate)	Fixed guaranteed bandwidth, minimal delay/jitter	Voice (VoIP), video conferencing
VBR (Variable Bit Rate)	Bandwidth varies but average/peak rate guaranteed	Compressed video, bursty transaction traffic

Service Category	Meaning	Typical Use
ABR (Available Bit Rate)	Uses feedback to share leftover bandwidth fairly	LAN interconnection, bulk data transfer
UBR (Unspecified Bit Rate)	Best-effort, no guarantee	Non-critical background traffic

5.2 Faults, Root Causes and Impact

Fault Observed	Root Cause	Impact
Delays in online transactions	Transaction traffic (VBR-appropriate) wrongly mapped to UBR, so it competes as best-effort under congestion	Slow banking response, poor customer experience
Voice jitter/packet loss	VoIP not assigned CBR category; cell delay variation (CDV) not bounded	Choppy or dropped calls
Reduced video quality	Video VCs share bandwidth with bursty data without VBR peak-rate enforcement	Frozen/pixelated video conferencing
ATM switches at high utilization	No traffic shaping/policing at ingress; traffic bursts exceed switch capacity at peak hours	Cell loss, increased queuing delay
Congested Virtual Circuits	Static VC bandwidth allocation not adjusted to actual demand; no traffic engineering	Cell loss and retransmission at higher layers
Inefficient bandwidth allocation	Connection Admission Control (CAC) thresholds not tuned; VCs over/under-provisioned	Some links idle while others saturated

6. Debugging Process

Step 1 — Baseline Utilization Check: Use SNMP polling / ATM switch CLI command 'show atm vc' and 'show interface atm utilization' to identify switches and VCs above 80% utilization, which indicates congestion hotspots.

Step 2 — Verify Service Category Mapping: Run 'show atm vc <vpi/vci>' on each switch to confirm which service category (CBR/VBR/ABR/UBR) is applied to voice, video and transaction VCs; misclassified categories are logged.

Step 3 — Measure Cell Loss Ratio (CLR) and Cell Delay Variation (CDV): Use ATM performance monitoring (OAM cells, RFC 1695 MIB) to quantify loss and jitter per VC and correlate with reported complaints.

Step 4 — Check Connection Admission Control (CAC) Logs: Review CAC rejection/acceptance logs to see if VCs were admitted without enough reserved bandwidth.

Step 5 — Analyze Traffic Shaping/Policing Configuration: Inspect ingress traffic contracts (PCR, SCR, MBS) to confirm if bursts are shaped correctly before entering the network.

Step 6 — Trace VC Path and Load Distribution: Use the ATM network management system to visualize VC routing and confirm whether traffic engineering is balancing load across available trunks.

Step 7 — Validate with Test Traffic: Generate synthetic CBR (voice-like) and VBR (video-like) test streams and measure end-to-end delay, CDV and CLR against ITU-T QoS targets.

7. Optimization Techniques

Technique	Explanation	Advantage	Limitation
Correct service-category remapping	Assign VoIP to CBR, video to VBR-rt, transactions to VBR-nrt, background data to UBR/ABR	Each traffic type gets appropriate QoS	Requires re-negotiating existing traffic contracts
Traffic shaping & policing at ingress	Enforce Peak Cell Rate (PCR) and Sustainable Cell Rate (SCR) using leaky-bucket algorithm	Prevents bursts from overwhelming switches	Adds minor processing overhead at ingress
Improved Connection Admission Control (CAC)	Tighten CAC to reject new VCs when bandwidth is insufficient, based on real utilization	Avoids over-subscription and cell loss	May reject some low-priority connection requests
ATM traffic engineering / VC re-routing	Redistribute VCs across multiple physical trunks instead of one congested path	Balances load, avoids single hotspot switch	Needs accurate topology and monitoring tools

Technique	Explanation	Advantage	Limitation
Bandwidth reservation tuning per VC	Recalculate SCR/PCR values based on actual measured demand rather than static defaults	Efficient use of available bandwidth	Requires periodic re-tuning as traffic grows

Comparison: the current system statically assigns bandwidth and often the wrong service category, causing congestion in some VCs while others sit idle. The optimized system dynamically shapes, polices and re-routes traffic based on real measured demand and correct QoS class, producing balanced utilization and predictable delay for every traffic type.

8. Network Diagram

```
   [HQ Data Center]                         [Regional Data Center]
          |                                          |
     [ATM Switch A] ===VC(CBR-Voice)=== [ATM Switch B] ===VC(CBR-Voice)=== [Regional Office]
          |  \==VC(VBR-Video)==\               |
          |                     \====VC(VBR-Video)====[ATM Switch C]===[Branch Office]
     [ATM Switch D]===VC(VBR-nrt Txn)===[ATM Switch B]
          |
   [Core Banking Servers]


   Legend: VC = Virtual Circuit, CBR = voice/video guaranteed, VBR = variable-rate
transaction/video
```

9. Flowchart — Troubleshooting Process

```
     [ Start: Performance Complaint ]
                |
                v
     [ Check Switch/VC Utilization (SNMP) ]
                |
     < Utilization > 80%? >-- Yes --> [ Investigate Congested Switch/VC ]
                | No                              |
                v                                 v
     [ Check Service-Category Mapping ]  [ Re-route VC / Add Trunk Capacity ]
                |
     < Wrong category assigned? >-- Yes --> [ Remap to correct CBR/VBR/ABR/UBR ]
                | No
                v
     [ Measure CLR and CDV via OAM cells ]
                |
```

```
< CLR/CDV exceeds ITU-T target? >-- Yes --> [ Apply Traffic Shaping / Tighten CAC ]
           | No
           v
[ Validate with Test Traffic ]
           |
           v
[ End: Document & Monitor Continuously ]
```

10. Algorithm / Procedure for Optimization

Step 1: Poll all ATM switches for VC-level utilization and identify congestion hotspots.

Step 2: Audit each VC's assigned service category against its actual traffic type.

Step 3: Remap voice VCs to CBR, real-time video to VBR-rt, transactions to VBR-nrt, bulk data to ABR/UBR.

Step 4: Configure ingress traffic shaping with correct PCR/SCR/MBS values using the leaky-bucket mechanism.

Step 5: Tighten CAC thresholds so new VCs are only admitted if bandwidth is genuinely available.

Step 6: Re-route or add capacity to congested trunks using traffic engineering.

Step 7: Send synthetic CBR/VBR test traffic and measure CLR, CDV and end-to-end delay.

Step 8: Compare results against ITU-T QoS targets; iterate configuration until targets are met.

Step 9: Deploy continuous OAM-based monitoring for early detection of future congestion.

11. Performance Analysis

Metric	Before Optimization	After Optimization	Expected Improvement
Transaction response time	3-6 seconds at peak	< 1 second	Faster banking transactions
Voice jitter (CDV)	40-80 ms	< 5 ms (CBR guaranteed)	Clear, stable voice calls
Cell Loss Ratio (CLR)	1-2%	< 0.01%	Near loss-free transmission
Switch utilization balance	Some switches > 90%, others < 30%	Balanced 50-70% across switches	Even load distribution
Video conferencing quality	Frequent freezing/pixelation	Smooth, guaranteed VBR-rt bandwidth	High-quality video calls

Metric	Before Optimization	After Optimization	Expected Improvement
Bandwidth utilization efficiency	Static, inefficient allocation	Dynamic, demand-based allocation	Better use of available capacity
Reliability during peak hours	Frequent congestion events	Congestion largely eliminated	Higher enterprise reliability

12. Security Improvements

12.1 Risks

Unauthenticated SVC setup requests could allow unauthorized VC creation; lack of encryption on ATM cells exposes sensitive financial data in transit; mismanaged CAC could be exploited to exhaust bandwidth (denial of service).

12.2 Solutions

Enable ATM security signaling standards to authenticate SVC setup requests; encrypt sensitive traffic at a higher layer (IPsec/TLS) before it enters the ATM network; enforce strict CAC and per-customer bandwidth caps to prevent resource exhaustion.

12.3 Best Practices

Maintain access control lists on switch management interfaces; log and audit all VC setup/teardown events; regularly patch ATM switch firmware; segregate management-plane traffic from data-plane traffic.

13. Testing and Validation

Test Case	Procedure	Expected Output	Result
TC1: CBR voice delay test	Send synthetic CBR voice stream across HQ-to-branch VC	CDV < 5 ms, CLR < 0.01%	Pass
TC2: VBR video load test	Stream test video during simulated peak load	No visible freezing, bandwidth honored	Pass
TC3: Transaction response test	Simulate 500 concurrent banking transactions	Response time < 1 s	Pass

Test Case	Procedure	Expected Output	Result
TC4: CAC admission test	Request new VC when link near capacity	Request correctly rejected/queued	Pass
TC5: Switch load balance test	Monitor utilization across all switches for 1 hour peak	All switches within 50-70% range	Pass

14. Advantages

Correct service-category mapping guarantees QoS per traffic type; traffic shaping and tightened CAC prevent congestion; traffic-engineered VC routing balances switch load; overall the bank gets faster transactions, clear voice calls and smooth video conferencing with high reliability.

15. Limitations

ATM is a legacy technology with high hardware cost and limited vendor support compared to modern IP/MPLS networks; re-provisioning VCs and CAC policies requires careful planning to avoid service interruption; ATM does not natively encrypt cells, so security still depends on higher-layer protocols.

16. Future Enhancements

Plan a phased migration from ATM to MPLS or IP/Ethernet-based WAN with equivalent or better QoS mechanisms (DiffServ, traffic engineering); introduce SDN-based centralized traffic engineering; add end-to-end encryption for all financial traffic; deploy AI-based predictive congestion management.

17. Conclusion

The ATM network's performance problems were caused by incorrect service-category mapping, absent traffic shaping, loose CAC policies and unbalanced VC routing. By correctly classifying voice into CBR, video into VBR-rt and transactions into VBR-nrt, applying traffic shaping and tightening admission control, and using traffic engineering to balance VC load, the bank achieves low-delay transactions, jitter-free voice calls and smooth video conferencing, restoring reliable enterprise-grade communication.

18. References (IEEE Format)

[1] ITU-T Recommendation I.150, "B-ISDN Asynchronous Transfer Mode Functional Characteristics," ITU-T, 1999.

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[4] B. A. Forouzan, Data Communications and Networking, 5th ed. New York, NY, USA: McGraw-Hill, 2013.

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SCENARIO 3: MPLS Network Debugging & Optimization

1. Aim

To debug and optimize a Multiprotocol Label Switching (MPLS) network connecting a multinational e-commerce company's headquarters, offices, warehouses and cloud data centers, ensuring fast, reliable performance for order processing, ERP, voice and video applications.

2. Problem Statement

The company's MPLS backbone carries order processing, ERP, VoIP, video conferencing and inventory traffic. During peak hours, cloud access is slow, orders are delayed, voice calls are poor, and video conferences are interrupted. Monitoring shows some Label Switched Paths (LSPs) are heavily congested while others are underused, some routers have incorrect label mappings causing inefficient routing, and traffic engineering has not been configured, leading to uneven bandwidth utilization.

3. Network Overview

MPLS (Multiprotocol Label Switching) is a technique that speeds up and controls the flow of network traffic by attaching short, fixed-length labels to packets instead of performing a full IP lookup at every router. Label Switch Routers (LSRs) forward packets based only on the label, following a pre-established Label Switched Path (LSP) from ingress to egress Label Edge Router (LER). This allows traffic engineering — administrators can force specific traffic classes onto specific paths to balance load and guarantee QoS. In this company, LERs sit at HQ, each regional office, each warehouse and the cloud data center edge, connected through core LSRs that switch labeled traffic across the WAN backbone.

4 Problem Identification

5.1 Faults, Root Causes and Impact

Fault Observed	Root Cause	Impact
Slow access to cloud applications	Cloud-bound LSPs congested; no traffic engineering to spread load across available paths	Delayed ERP and inventory queries
Delayed order processing	Order traffic sharing the same congested LSP as bulk/background traffic, no per-class prioritization	Slower checkout, customer dissatisfaction

Fault Observed	Root Cause	Impact
Poor-quality voice calls	VoIP LSP not marked with correct EXP/DSCP bits, so it is treated as best-effort under congestion	Jitter and dropped calls
Video conference interruptions	Video traffic sharing congested LSP without bandwidth reservation (RSVP-TE)	Frozen or dropped video sessions
Some LSPs heavily congested, others idle	Static LSP setup without traffic-engineering (CSPF-based path computation); default IGP shortest path used for all traffic	Uneven backbone utilization, wasted capacity
Incorrect label mappings on routers	Manual/legacy Label Information Base (LIB) misconfiguration or stale LDP bindings after topology changes	Packets forwarded on wrong path or dropped, increased latency
Uneven bandwidth utilization	No traffic-engineering tunnels (TE-LSPs); all traffic follows IGP metric, ignoring actual link load	Congestion on shortest-path links while alternate links unused

5. Debugging Process

Step 1 — Verify LSP Status: Run 'show mpls ldp neighbor' and 'show mpls forwarding-table' on core routers to confirm LSPs are correctly established end to end. This detects broken or missing LSPs first.

Step 2 — Check Label Bindings: Run 'show mpls ldp bindings' to compare local and remote label mappings across routers and identify stale or incorrect bindings after any topology change.

Step 3 — Measure Per-LSP Utilization: Use MPLS-aware SNMP/NetFlow monitoring to plot bandwidth usage per LSP and identify which LSPs exceed 80% utilization versus which are near-idle.

Step 4 — Inspect QoS Marking: Check whether packets are correctly marked with MPLS EXP bits (or DSCP mapped to EXP) for voice, video, order-processing and bulk traffic classes using 'show mpls experimental' style commands.

Step 5 — Trace Path with MPLS OAM Tools: Use MPLS ping and MPLS traceroute (LSP ping/traceroute per RFC 4379) to verify the actual path taken by an LSP matches the intended path and to detect black-holed labels.

Step 6 — Review Traffic-Engineering Configuration: Check whether RSVP-TE or CSPF-based TE-tunnels are configured; if all traffic is using default IGP shortest path, this confirms the missing-traffic-engineering root cause.

Step 7 — Validate with Synthetic Load: Generate test order-processing, voice and video traffic and measure latency, jitter, and loss across the identified LSPs to quantify the improvement needed.

6. Optimization Techniques

Technique	Explanation	Advantage	Limitation
Traffic Engineering with RSVP-TE tunnels	Explicitly compute and reserve paths (CSPF) based on real bandwidth availability instead of plain IGP shortest path	Balances load, avoids congestion hotspots	Adds signaling overhead and configuration complexity
Correct label/LDP re-synchronization	Clear and rebuild LDP sessions/bindings after topology change; verify with LSP ping	Removes black-holed or mis-routed traffic	Brief interruption during re-sync
MPLS QoS with DiffServ-TE	Mark and prioritize voice (EF), video (AF41), order-processing (AF31) and bulk (BE) using EXP bits, with class-based bandwidth reservation per LSP	Each traffic class gets guaranteed treatment	Requires consistent marking policy across all edge routers
Fast Reroute (FRR) for LSPs	Pre-computed backup LSPs activate within tens of milliseconds if the primary path fails	High availability, minimal disruption	Requires extra reserved backup bandwidth
Load balancing across ECMP/TE paths	Distribute traffic across multiple equal/unequal cost LSPs based on measured link utilization	Efficient use of all backbone capacity	Needs continuous monitoring and re-optimization

Comparison: currently all traffic blindly follows the IGP shortest path with no bandwidth awareness, so a few links become bottlenecks while others are wasted, and mis-synchronized labels silently drop or misroute packets. After optimization, RSVP-TE tunnels place traffic on paths chosen by actual available bandwidth, DiffServ-TE guarantees per-class treatment, and Fast Reroute protects against failures, resulting in balanced, predictable, and resilient performance.

7. Network Diagram

```
  [HQ - LER1]                                        [Cloud DC - LER4]
       \                                                   /
```

```
      \==LSP1(TE, Order/ERP, AF31)==[LSR-Core1]==LSP1==/
       \                                |
        \==LSP2(TE, VoIP/Video, EF/AF41)=[LSR-Core2]
                                         |
[Warehouse - LER2]==LSP3(Inventory)======|
                                         |
[Regional Office - LER3]==LSP4(Backup/FRR)=[LSR-Core1]


LER = Label Edge Router,  LSR = Label Switch Router, LSP = Label Switched Path
```

8. Flowchart — Troubleshooting Process

```
  [ Start: Slow/Poor Quality Reported ]
              |
              v
  [ Verify LSP Status (show mpls forwarding-table) ]
              |
   < LSP down/missing? >-- Yes --> [ Rebuild LDP Session ]
              | No
              v
  [ Check Label Bindings for Mismatch ]
              |
   < Stale/incorrect binding? >-- Yes --> [ Clear & Re-sync LDP Bindings ]
              | No
              v
  [ Measure Per-LSP Utilization ]
              |
   < LSP > 80% utilized? >-- Yes --> [ Configure RSVP-TE / Reroute Traffic ]
              | No
              v
  [ Check QoS/EXP Marking ]
              |
   < Marking missing/incorrect? >-- Yes --> [ Apply DiffServ-TE Class Marking ]
              | No
              v
  [ LSP Ping/Traceroute Validation ]
              |
              v
  [ End: Document & Continuous Monitoring ]
```

9 Algorithm / Procedure for Optimization

Step 1: Verify every LSP is up using MPLS ping/traceroute and forwarding-table checks.

Step 2: Audit and correct any stale or mismatched label bindings via LDP re-synchronization.

Step 3: Measure per-LSP bandwidth utilization and flag LSPs above threshold (e.g., 80%).

Step 4: Design RSVP-TE tunnels with CSPF path computation based on real available bandwidth.

Step 5: Apply DiffServ-TE class marking: EF for voice, AF41 for video, AF31 for order/ERP, BE for bulk.

Step 6: Configure Fast Reroute backup LSPs for critical traffic classes.

Step 7: Redistribute traffic across underused LSPs using explicit-path or affinity constraints.

Step 8: Send synthetic test traffic per class and validate latency, jitter and loss against targets.

Step 9: Enable continuous MPLS-aware monitoring (NetFlow/SNMP) to catch future imbalance early.

10. Performance Analysis

Metric	Before Optimization	After Optimization	Expected Improvement
Order processing latency	2-5 seconds at peak	< 500 ms	Faster checkout experience
Cloud/ERP access delay	High, inconsistent	Low, consistent	Reliable ERP performance
Voice call jitter	50-100 ms	< 10 ms (EF class)	Clear VoIP calls
Video conference stability	Frequent freezing/drop	Stable with reserved bandwidth	Smooth video sessions
LSP utilization balance	Some LSPs > 90%, others < 20%	Balanced 40-65% via TE	Efficient backbone usage
Packet loss due to label errors	Noticeable due to stale bindings	Near zero after LDP re-sync	Accurate label forwarding
Failover time on link failure	Seconds (IGP reconvergence)	< 50 ms (Fast Reroute)	High resilience

11. Security Improvements

11.1 Risks

MPLS by itself does not encrypt payloads, so sensitive order and customer data could be exposed if the WAN is compromised; unauthorized label injection could redirect traffic; misconfigured VPN route targets could leak routes between customer VPNs (VRF leakage).

11.2 Solutions

Layer IPsec encryption over MPLS VPN links for sensitive traffic; strictly control route-target import/export policies in MPLS L3VPN configuration to prevent VRF leakage; authenticate LDP sessions using MD5/TCP-AO to prevent label-spoofing attacks.

11.3 Best Practices

Regularly audit VRF route-target configurations; restrict router management access with ACLs and AAA; monitor for unexpected label bindings; keep router OS patched against known MPLS vulnerabilities.

12. Testing and Validation

Test Case	Procedure	Expected Output	Result
TC1: LSP integrity test	Run MPLS LSP ping on all core LSPs	100% reachability, no black-holing	Pass
TC2: TE load balance test	Monitor per-LSP utilization for 1-hour peak window	All LSPs within 40-65% range	Pass
TC3: Voice QoS test	Generate EF-marked VoIP stream under load	Jitter < 10 ms, loss < 0.1%	Pass
TC4: Order processing load test	Simulate 1000 concurrent checkout transactions	Latency < 500 ms	Pass
TC5: Fast Reroute failover test	Simulate core link failure during active session	Failover < 50 ms, no session drop	Pass

13. Advantages

Traffic engineering removes congestion hotspots and uses backbone capacity efficiently; DiffServ-TE guarantees QoS for voice, video and order-processing; Fast Reroute provides sub-50-ms failover for high availability; corrected label bindings eliminate silent packet loss.

14. Limitations

RSVP-TE and DiffServ-TE configuration is complex and needs skilled staff to maintain; reserved bandwidth for FRR backups reduces capacity available for normal traffic; MPLS itself does not provide encryption, so additional security layers are still required.

15. Future Enhancements

Migrate toward Segment Routing (SR-MPLS) to simplify traffic engineering without RSVP signaling overhead; introduce SDN-based centralized controller for real-time traffic-engineering decisions; integrate AI/ML-based traffic prediction for proactive LSP re-optimization before congestion occurs.

16. Conclusion

The MPLS network's performance issues were rooted in the absence of traffic engineering, incorrect label bindings, and missing QoS marking, causing some LSPs to be congested while others sat idle. By verifying and correcting label bindings, deploying RSVP-TE traffic engineering with DiffServ-TE class marking, and adding Fast Reroute for resilience, the e-commerce company's backbone now delivers fast order processing, reliable ERP access, clear voice calls and stable video conferencing.

17. References (IEEE Format)

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SCENARIO 4: Error Control Debugging & Optimization in an Automated Warehouse Network

2. Aim

To investigate transmission errors in an automated warehouse network (barcode scanners, RFID readers, robots and inventory systems), identify weaknesses in the existing error-detection/correction mechanisms, and recommend an optimized error-control strategy for accurate, reliable data communication.

3. Problem Statement

In the automated warehouse, barcode scanners, RFID readers, autonomous robots and inventory systems continuously exchange shipment and inventory data with a central server. Some shipment records arrive with incorrect information, barcode scans occasionally fail, and inventory counts mismatch physical stock. Peak-hour monitoring shows increased transmission errors from electrical interference, signal attenuation and congestion; some corrupted frames are detected and retransmitted, but others go undetected, causing wrong database entries and delayed orders. The current error-detection/correction configuration must be debugged and optimized.

4. Network Overview

Error control is the set of techniques used at the Data Link Layer (and sometimes higher layers) to detect and, where possible, correct errors introduced during transmission by noise, interference or attenuation. Common error-detection methods include Parity Check (simple, detects single-bit errors), Checksum (used in TCP/UDP/IP headers), and Cyclic Redundancy Check or CRC (strong, used in Ethernet frames). Error-correction methods such as Hamming Code or Forward Error Correction (FEC) allow the receiver to fix certain errors without asking for retransmission. In the warehouse, RFID readers and barcode scanners send short data frames over a mix of wireless (Wi-Fi/RFID RF links) and wired Ethernet to the central inventory server, where every bit of shipment/inventory data must be received correctly.

5. Problem Identification

5.1 Faults, Root Causes and Impact

Fault Observed	Root Cause	Impact
Shipment records with incorrect information	Weak error-detection code (e.g., simple parity) unable to catch multi-bit or burst errors	Wrong data silently accepted into database
Barcode scans occasionally fail	RF/optical signal attenuation and electrical interference from warehouse machinery (motors, conveyor belts)	Failed reads, need for repeated scanning, slower operations
Inventory count mismatches	Undetected errors corrupt quantity fields without triggering retransmission	Stock records diverge from physical stock
Increased transmission errors during peak hours	Higher device density and RF/electrical noise when more robots/scanners operate simultaneously	Higher Bit Error Rate (BER) at busy times
Some corrupted frames undetected	Error-detection scheme too weak for burst errors (e.g., single-bit parity cannot catch even-numbered bit flips or burst noise)	Silent data corruption in inventory database
Corrected frames retransmitted but delay order processing	Automatic Repeat reQuest (ARQ) retransmission without error correction adds round-trip delay for every detected error	Slower processing pipeline during high load

6. Debugging Process

Step 1 — Measure Bit Error Rate (BER): Use a protocol analyzer or built-in NIC statistics to record BER on wired Ethernet links and RF signal quality tools for RFID/Wi-Fi links, focusing on peak-hour periods.

Step 2 — Identify the Current Error-Detection Scheme: Inspect device/firmware configuration to confirm whether frames use simple parity, checksum, or CRC, since this determines detection strength.

Step 3 — Correlate Errors with Interference Sources: Use a spectrum analyzer near conveyor motors and RFID readers to detect electromagnetic interference (EMI) coinciding with error spikes.

Step 4 — Check Frame-Level Logs for Silent Errors: Compare application-level checksums (e.g., a hash of the shipment record) against what was stored, to find cases where link-layer error checking passed but data was still wrong — proving the detection code is too weak.

Step 5 — Analyze Retransmission (ARQ) Statistics: Check retransmission counters at the data-link layer to see how often ARQ is triggered and how much delay it adds during peak load.

Step 6 — Physical-Layer Inspection: Inspect cabling, connectors, and RFID antenna placement for attenuation or damage contributing to weak signal strength.

Step 7 — Validate Root Cause with Controlled Test: Deploy a test scanner near a known interference source and log error rate, then move it away and re-measure, confirming EMI as a contributing cause.

7. Optimization Techniques

Technique	Explanation	Advantage	Limitation
Upgrade to CRC-32 for frame error detection	Replace simple parity/checksum with Cyclic Redundancy Check, which reliably detects burst errors and most multi-bit errors	Much stronger error detection than parity	Slightly higher computation overhead per frame
Forward Error Correction (FEC) / Hamming Code	Add redundant bits so the receiver can correct certain errors without requesting retransmission	Reduces retransmission delay, works well for real-time flow	Adds bandwidth overhead for redundant bits
Hybrid ARQ (combine FEC + selective-repeat ARQ)	Use FEC to fix minor errors instantly; fall back to selective-repeat ARQ retransmission only for uncorrectable frames	Best balance of speed and reliability	More complex to implement than plain ARQ
Physical-layer interference mitigation	Add shielding/filtering near motors, reposition RFID antennas, use twisted-pair/fiber where interference is high	Reduces root-cause noise, fewer errors overall	May require hardware changes/cost
Application-level end-to-end checksum/hash	Add a hash (e.g., CRC or SHA checksum) of the full shipment record verified at the server before committing to database	Catches any error that slipped past lower layers	Adds minor processing at server and device

Comparison: the current system relies on a weak detection code that misses burst errors, so corrupted data can enter the database silently, and retransmission-only ARQ adds delay for every detected error. The optimized system combines strong CRC detection, Hamming/FEC correction for common single-bit errors, hybrid ARQ retransmission only when necessary, and an application-level integrity check as a final safety net, plus physical-layer interference reduction to lower the error rate at its source.

8. Network Diagram

```
  [Barcode Scanner]   [RFID Reader]   [Autonomous Robot]
         \                 |                  /
          \                |                 /
       (Wireless/RF Links -- source of noise/attenuation)
                            |
                  [Warehouse Access Point / Gateway]
                            |
                  [Wired Ethernet Backbone] --(EMI from motors/conveyors)
                            |
                 [Central Inventory Server / Database]
                            |
                 [ Order Processing System ]
```

9. Flowchart — Troubleshooting Process

```
    [ Start: Data Mismatch/Scan Failure Reported ]
                   |
                   v
    [ Measure Bit Error Rate (BER) at Peak Hours ]
                   |
    < BER above threshold? >-- Yes --> [ Identify Interference Source (Spectrum Analyzer) ]
                   | No                            |
                   v                                v
    [ Check Error-Detection Scheme Strength ]  [ Apply Shielding / Reposition Antennas ]
                   |
    < Using weak parity/checksum? >-- Yes --> [ Upgrade to CRC-32 ]
                   | No
                   v
    [ Check for Silent Errors (App-level hash mismatch) ]
                   |
    < Silent errors found? >-- Yes --> [ Add FEC/Hamming Code + End-to-End Checksum ]
                   | No
                   v
    [ Analyze ARQ Retransmission Delay ]
                   |
    < Excess delay? >-- Yes --> [ Implement Hybrid ARQ (FEC + Selective Repeat) ]
                   | No
```

```
                  v
    [ End: Validate & Monitor Continuously ]
```

10. Algorithm / Procedure for Optimization

Step 1: Measure baseline BER and correlate error spikes with time of day and nearby machinery.

Step 2: Identify and document the currently used error-detection scheme on each device type.

Step 3: Replace simple parity/checksum with CRC-32 at the data-link layer for all scanner/RFID/robot traffic.

Step 4: Add Hamming Code or FEC redundancy for single-bit correction on the most error-prone RF links.

Step 5: Implement Hybrid ARQ so only uncorrectable frames trigger selective-repeat retransmission.

Step 6: Add an application-level end-to-end checksum verified before committing any record to the database.

Step 7: Reduce physical interference: shield cabling, reposition RFID antennas away from motors, use fiber where feasible.

Step 8: Re-measure BER, silent-error rate and retransmission delay against baseline; iterate until targets are met.

Step 9: Deploy continuous monitoring dashboards to alert if BER rises above threshold in future.

11. Performance Analysis

Metric	Before Optimization	After Optimization	Expected Improvement
Bit Error Rate (BER)	1 in 10^4 bits (peak hours)	1 in 10^7 bits or better	Far fewer transmission errors
Undetected (silent) errors	Occasional, causing DB mismatches	Near zero (CRC + app-level hash)	Accurate database records
Retransmission-caused delay	High, every detected error re-sent	Reduced (FEC corrects most errors instantly)	Faster order processing
Barcode scan failure rate	Noticeable during peak load	Significantly reduced	Fewer repeated scans

Metric	Before Optimization	After Optimization	Expected Improvement
Inventory accuracy	Mismatches between DB and physical stock	DB matches physical stock closely	Reliable stock management
Network reliability under interference	Degrades sharply near machinery	Stable due to shielding + FEC	Consistent performance warehouse-wide

12. Security Improvements

12.1 Risks

Unencrypted RFID/barcode data could be intercepted or spoofed (fake tag injection); weak error-checking could let maliciously altered frames pass undetected; open Wi-Fi networks used by scanners could be accessed by unauthorized devices.

12.2 Solutions

Encrypt data between RFID readers/scanners and the server using WPA3/TLS as applicable; combine strong CRC/FEC with cryptographic integrity checks (e.g., HMAC) so tampering is detected, not just random noise; place warehouse IoT devices on an isolated, authenticated VLAN.

12.3 Best Practices

Regularly update scanner/RFID firmware; use unique device credentials instead of shared keys; monitor for unusual device behavior (possible spoofed tags); physically secure network equipment in the warehouse.

13. Testing and Validation

Test Case	Procedure	Expected Output	Result
TC1: CRC error-detection test	Inject known bit errors into test frames near motors	All injected errors detected	Pass
TC2: FEC correction test	Introduce single-bit errors on RF link	Frame auto-corrected without retransmission	Pass
TC3: Silent-error check	Compare app-level hash of 10,000 records before/after transmission	Zero mismatches	Pass

Test Case	Procedure	Expected Output	Result
TC4: Peak-hour BER test	Measure BER during simulated peak robot/scanner activity	BER ≤ target threshold	Pass
TC5: Retransmission delay test	Measure average processing delay under Hybrid ARQ vs old ARQ	Delay reduced significantly	Pass

14. Advantages

CRC-32 catches burst errors that simple parity misses; FEC/Hamming code corrects common errors instantly, cutting delay; the application-level checksum acts as a final safety net against silent corruption; physical-layer fixes reduce the root cause of noise, improving overall reliability of the automated warehouse.

15. Limitations

FEC adds bandwidth overhead due to redundant bits; CRC and Hamming code require additional processing at every device, which may need firmware/hardware upgrades on older scanners; physical shielding and antenna repositioning involve installation cost and warehouse downtime.

16. Future Enhancements

Move toward more advanced FEC schemes (e.g., Reed-Solomon codes) for higher correction capability; deploy edge computing at the warehouse to validate data locally before it reaches the central server; use AI-based anomaly detection to flag suspicious inventory changes automatically; migrate wireless links to more interference-resistant technologies (e.g., Wi-Fi 6 with better modulation).

17. Conclusion

The warehouse's data errors were caused by a weak parity/checksum error-detection scheme, unmitigated electrical/RF interference, and a slow retransmission-only recovery method, together resulting in silent database corruption and delayed processing. By upgrading to CRC-32 detection, adding FEC/Hamming correction, adopting Hybrid ARQ, layering an application-level integrity check, and reducing physical interference, the network now achieves a much lower error rate, near-zero silent corruption, and faster, more accurate inventory and shipment processing

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SCENARIO 5: Flow Control Debugging & Optimization for an Online Examination Platform

2. Aim

To investigate the flow-control problems of an online education platform's live-class and examination system, debug the causes of buffer overflow, retransmissions and poor throughput, and recommend optimization techniques for reliable, efficient data transmission during high-traffic periods.

3. Problem Statement

An online education platform with 15,000+ students conducts live classes and exams where a 10 Gbps central server simultaneously sends question papers, images, videos and acknowledgements to students on much slower broadband/mobile connections. During peak hours, students experience incomplete downloads, delayed page loads, duplicate packets and repeated retransmissions. Some students receive new data before finishing processing the previous data, causing receiver buffer overflow and application crashes. Administrators observe excessive retransmissions, high latency, reduced throughput and poor bandwidth utilization because the current flow-control mechanism does not properly regulate the sender's transmission rate to match each receiver's capacity.

4. Network Overview

Flow control is a Transport/Data-Link Layer mechanism that prevents a fast sender from overwhelming a slow receiver by regulating how much data can be sent before an acknowledgement is required. Common flow-control techniques include Stop-and-Wait (sender waits for ACK after every frame — simple but slow), Sliding Window (sender can send multiple frames up to a window size before waiting), and TCP's dynamic Receive Window combined with Congestion Control (window size is adjusted based on receiver buffer space and network congestion). In this platform, the central exam server is the sender with very high bandwidth (10 Gbps), while each student's device is a receiver with limited bandwidth and buffer size — a classic fast-sender/slow-receiver mismatch that flow control exists specifically to solve.

5. Problem Identification

5.1 Faults, Root Causes and Impact

Fault Observed	Root Cause	Impact
Incomplete downloads / delayed page loading	Server sends data faster than the student's slow broadband/mobile link	Students cannot access question papers or videos in time

Fault Observed	Root Cause	Impact
	can absorb, without adjusting to per-client bandwidth	
Duplicate packets and repeated retransmissions	Timeout-based retransmission fires before the original packet's ACK arrives (timer too short for slow links), causing needless re-sends	Wasted bandwidth, more congestion
Receiver buffer overflow / application crash	Server keeps pushing new data before the client finishes processing previous data; fixed/large send window not adapted to small receiver buffer	Data loss, app crash, failed exam submission
Excessive retransmissions	Sliding window size too large for low-bandwidth clients; no dynamic window scaling	Increased latency, wasted server bandwidth
Increased latency, reduced throughput	No differentiation between fast and slow clients — same transmission strategy applied uniformly to all 15,000 students	Uneven experience; fast clients wasted, slow clients overwhelmed
Poor bandwidth utilization at server	No client-side buffer feedback used to throttle sender; missing adaptive flow control	10 Gbps server capacity not effectively delivered to end users

6. Debugging Process

Step 1 — Capture Traffic with Wireshark: Record a live packet capture during a simulated exam session to inspect TCP window size, retransmission count and duplicate ACKs for a slow client.

Step 2 — Analyze TCP Window Behavior: Check whether the TCP Receive Window (rwnd) advertised by slow clients is being respected by the server, or whether the server continues sending regardless — revealing whether flow control is actually functioning.

Step 3 — Check Retransmission Timeout (RTO) Settings: Compare the configured RTO against actual measured Round-Trip Time (RTT) for slow mobile clients; an RTO shorter than real RTT causes unnecessary retransmissions.

Step 4 — Monitor Server-Side Buffer and Send Rate: Use server monitoring tools to check if the send buffer size and pacing are static (one-size-fits-all) rather than adapted per connection.

Step 5 — Test with Simulated Slow Client: Throttle a test client's bandwidth (e.g., using tc/netem) to emulate mobile broadband and record buffer-overflow behavior, confirming the fast-sender/slow-receiver mismatch.

Step 6 — Review Application-Layer Flow Control: Check whether the exam application itself paces content delivery (e.g., chunked question loading) or relies solely on the transport layer.

Step 7 — Validate Findings Against Peak-Hour Logs: Cross-check packet-capture findings with server logs from an actual peak-hour exam session to confirm the debugging conclusions at scale.

7. Optimization Techniques

Technique	Explanation	Advantage	Limitation
Dynamic Sliding Window with TCP Window Scaling	Let each client's advertised receive window (with RFC 1323 scaling) genuinely govern how much unacknowledged data the server sends	Server automatically adapts to each client's real capacity	Requires OS/network support for window scaling on both ends
Adaptive RTO calculation (Karn's/Jacobson's algorithm)	Compute retransmission timeout from measured RTT and its variance instead of a fixed short value	Reduces unnecessary duplicate retransmissions	Needs continuous RTT sampling per connection
Server-side rate limiting / traffic shaping per client class	Group clients by measured bandwidth (e.g., broadband vs mobile) and pace delivery accordingly	Prevents overwhelming slow clients while not throttling fast ones	Requires client bandwidth classification logic
Content chunking with client-paced pull (HTTP range requests / adaptive streaming)	Let the client request the next chunk (question, image, video segment) only after processing the previous one, instead of server push	Client naturally controls pace, buffer overflow avoided	Slightly higher request overhead per chunk
CDN / edge caching for static exam content	Serve images, question papers and video from geographically	Reduces load on central server and latency for students	Additional infrastructure cost

Technique	Explanation	Advantage	Limitation
	distributed edge servers close to students		
Load balancing across multiple exam servers	Distribute 15,000 students across several servers instead of one, reducing per-server send pressure	Better overall throughput and resilience	Requires session synchronization across servers

Comparison: the current design pushes data from a single fast 10 Gbps server uniformly to all clients regardless of their real capacity, causing slow clients' buffers to overflow and triggering wasteful retransmissions. The optimized design lets each client's true receive window and RTT drive the server's send rate, paces delivery through client-pulled chunking, and spreads load using CDN/edge caching and multiple servers — so every student, regardless of connection speed, receives data at a pace their device and link can actually handle.

8. Network Diagram

```
                    [ Central Exam Server - 10 Gbps ]
                                 |
                     [ Load Balancer / CDN Edge ]
          ______________________|______________________
         |                       |                        |
   [Edge Node - Region A] [Edge Node - Region B]  [Edge Node - Region C]
         |                       |                        |
 (Broadband Students)   (Mobile 4G Students)     (Broadband Students)
 High rwnd, fast ACK    Small rwnd, slow ACK      High rwnd, fast ACK

 Flow control: each client's TCP receive-window (rwnd) throttles
 how fast the server/edge node can send new question/video data.
```

9. Flowchart — Troubleshooting Process

```
   [ Start: Buffer Overflow / Retransmission Complaint ]
                   |
                   v
   [ Capture Traffic (Wireshark) During Peak Load ]
                   |
   < Server ignoring client rwnd? >-- Yes --> [ Enable/Enforce TCP Window Scaling ]
                   | No
                   v
   [ Compare RTO vs Measured RTT ]
```

```
                |
< RTO too short for slow clients? >-- Yes --> [ Apply Adaptive RTO (Jacobson's Algorithm) ]
                | No
                v
[ Check Server Send Pattern (Push vs Pull) ]
                |
< Server pushing blindly? >-- Yes --> [ Switch to Client-Paced Chunked Delivery ]
                | No
                v
[ Check Load Distribution Across Servers ]
                |
< Single server overloaded? >-- Yes --> [ Add CDN/Edge Caching + Load Balancing ]
                | No
                v
[ End: Validate with Simulated Peak-Load Test ]
```

10. Algorithm / Procedure for Optimization

Step 1: Capture and analyze TCP flow behavior for a representative sample of slow and fast clients.

Step 2: Enable/verify TCP window scaling so each client's real receive window governs the server's outstanding data.

Step 3: Replace fixed RTO with adaptive RTO computed from sampled RTT and variance (Jacobson's algorithm).

Step 4: Classify clients by measured bandwidth and apply server-side rate shaping per class.

Step 5: Redesign content delivery as client-paced chunked pull instead of server push.

Step 6: Deploy CDN/edge caching for static content (question images, pre-recorded videos).

Step 7: Load-balance the 15,000 students across multiple exam servers/edge nodes.

Step 8: Run a simulated peak-load test with mixed fast/slow clients and measure throughput, retransmissions and buffer-overflow incidents.

Step 9: Iterate configuration until targets are met, then monitor continuously during live exams.

11. Performance Analysis

Metric	Before Optimization	After Optimization	Expected Improvement
Retransmission rate	High (10-20% of packets)	Low (<2%)	Less wasted bandwidth
Receiver buffer overflow incidents	Frequent among mobile/slow clients	Rare to none	Fewer app crashes
Throughput per slow client	Very low, inconsistent	Matches client's real link capacity	Smooth, complete downloads
Average latency	High during peak hours	Significantly reduced	Faster page/question loading
Server bandwidth utilization	Poor (10 Gbps underused effectively)	Efficient, load-balanced across edges	Better use of server investment
Duplicate packets	Common due to premature retransmission	Minimized via adaptive RTO	Cleaner network traffic
Scalability (more students)	Degrades sharply with more students	Scales via CDN/load balancing	Supports growth beyond 15,000 users

12. Security Improvements

12.1 Risks

Exam content delivered without proper session validation could be intercepted or replayed; overloaded servers are more vulnerable to being mistaken for or exploited as a denial-of-service condition; unauthenticated chunk-pull requests could be abused to scrape exam content.

12.2 Solutions

Use HTTPS/TLS for all question, image and video delivery; implement per-session tokens so chunk-pull requests are authenticated and rate-limited per student; deploy DDoS protection at the CDN/edge layer given the large simultaneous user base.

12.3 Best Practices

Rotate exam session tokens; log and monitor abnormal request patterns (possible scraping); ensure CDN edge nodes enforce the same access-control policy as the origin server; regularly load-test before major exam events.

13. Testing and Validation

Test Case	Procedure	Expected Output	Result
TC1: Slow-client throughput test	Throttle test client to mobile-broadband speed, download full exam content	Complete download, no buffer overflow	Pass
TC2: Adaptive RTO test	Measure retransmissions before/after Jacobson's RTO on high-latency link	Retransmissions reduced significantly	Pass
TC3: Peak-load simulation	Simulate 15,000 concurrent client sessions	Stable throughput, latency within target	Pass
TC4: CDN edge delivery test	Request static content from 3 different regions	Content served from nearest edge, low latency	Pass
TC5: Buffer overflow stress test	Push data faster than a deliberately slow client can consume	Flow control throttles sender, no crash	Pass

14. Advantages

Dynamic window-based flow control matches server output to each client's real capacity, eliminating buffer overflow; adaptive RTO removes wasteful duplicate retransmissions; CDN/edge caching plus load balancing lets the platform scale smoothly to large numbers of simultaneous exam-takers with consistent performance.

15. Limitations

Client-paced chunked delivery adds slightly more request overhead compared to simple push; CDN and multi-server load balancing require additional infrastructure investment and session-synchronization complexity; performance still ultimately depends on each student's own last-mile internet quality, which the platform cannot fully control.

16. Future Enhancements

Adopt QUIC/HTTP-3 for improved flow control and faster connection setup on lossy mobile links; use machine-learning-based bandwidth prediction to pre-adjust delivery rate per student before congestion occurs; provide an offline-capable exam mode that pre-downloads content and syncs answers when connectivity is poor.

17. Conclusion

The platform's flow-control failures stemmed from a high-speed server pushing data uniformly to clients of widely varying capacity, an RTO too short for slow mobile links, and a single-server architecture unable to scale to 15,000 concurrent users. By enabling genuine TCP window-scaling-based flow control, adopting adaptive RTO, switching to client-paced chunked delivery, and distributing load through CDN edge caching and multiple servers, the platform now delivers exam content reliably and efficiently to every student regardless of connection speed, eliminating buffer overflow and application crashes.

19. References (IEEE Format)

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